Senior *Web3* Backend Engineer — *Web3* Fintech & Payments

Location: Remote / Poland (International Team)
Employment Type: Full-time / Part-time / Contract
Reports To: Backend Engineering Lead / Head of Engineering
Level: Senior Engineer (Hands-on Technical Role)

About the Company

Plaid is a financial technology company that provides infrastructure connecting financial applications with banks and financial institutions. Its technology supports secure financial data access, account verification, identity verification, and payment services.

Separately, we are planning to establish a new software development company in Poland, focusing on fintech, payment infrastructure, and *blockchain*-based financial products.

Our planned projects cover real estate marketplaces, gaming, sports betting, and *Web3* social finance platforms. We aim to combine traditional financial technologies with *blockchain* infrastructure to support digital payments, *wallet* integrations, transaction processing, *smart contract*s, and decentralized financial services.

We are building an international engineering team and looking for experienced engineers who can contribute to the development of secure, scalable, and reliable financial applications from an early stage.

About the Role

We are looking for an experienced **Senior *Web3* Backend Engineer** to design, develop, and maintain backend systems supporting our fintech, payments, and *blockchain*-connected applications.

This is a hands-on engineering role focused on backend architecture, API development, distributed systems, *blockchain* integrations, and reliable financial transaction processing.

You will work closely with *blockchain* engineers, *smart contract* developers, frontend engineers, DevOps engineers, and product teams to build the services connecting traditional financial applications with *blockchain* networks.

Your responsibilities will include developing scalable backend APIs, implementing transaction-processing workflows, integrating *smart contract*s, and maintaining reliable communication between off-chain services and on-chain infrastructure.

We are looking for someone with s*tron*g backend engineering fundamentals who understands the practical challenges of building production *Web3* applications.

This is not a purely *smart contract* development or *blockchain* protocol engineering position. The primary focus is building secure and reliable backend systems that support financial transactions and *blockchain* integrations.

What You'll Do

Backend Development & Architecture

- Design, develop, and maintain scalable backend services using **Node.js, TypeScript, and NestJS**.
- Build secure REST APIs, GraphQL services, and internal service integrations.
- Develop modular backend architectures that support multiple fintech and *Web3* products.
- Design microservices and event-driven systems where appropriate.
- Implement asynchronous processing, background jobs, queues, and scheduled tasks.
- Build reliable systems for processing financial and *blockchain*-related transactions.
- Design database schemas, improve query performance, and implement effective caching strategies.
- Work with PostgreSQL, Redis, and other databases according to application requirements.
- Implement authentication, authorization, access controls, and secure API communication.
- Improve backend performance, scalability, reliability, and maintainability.
- Participate in system design discussions and contribute to important architecture decisions.
- Take ownership of backend features from technical design through deployment and production maintenance.

Web3 & *Blockchain* Integrations

- Develop backend services that communicate with *Ethereum* and *EVM*-compatible *blockchain* networks.
- Integrate *smart contract*s using Viem, *Eth*ers.js, *Web3*.js, or similar *blockchain* libraries.
- Build services for reading *blockchain* state, monitoring contract events, and processing on-chain data.
- Implement *blockchain* transaction submission, confirmation tracking, and error handling.

- Develop *wallet*-connected backend functionality supporting digital assets and financial applications.
- Build services that track deposits, withdrawals, *token* transfers, and *blockchain* payment activity.
- Support *blockchain* in*dex*ing, transaction monitoring, and event-driven synchronization.
- Implement reliable handling for failed transactions, delayed confirmations, RPC failures, and chain reorganizations.
- Collaborate with *blockchain* engineers to ensure correct *smart contract* integrations.
- Support integrations with multiple *blockchain* networks when required by product needs.
- Evaluate RPC providers and *blockchain* infrastructure solutions based on reliability, performance, and cost.

Fintech & Payments Infrastructure

- Build backend services supporting payments, financial transactions, and digital asset operations.
- Develop transaction-processing workflows for deposits, withdrawals, transfers, refunds, and settlement.
- Integrate traditional payment APIs and *blockchain* transaction services.
- Implement transaction validation, idempotency, retries, and error recovery mechanisms.
- Maintain consistent financial transaction states across multiple services.
- Design reliable reconciliation processes between application records and external payment or *blockchain* systems.
- Develop payment status tracking and transaction history services.
- Support secure integrations with *wallet* infrastructure, identity verification, and KYC providers where required.
- Collaborate with product and security teams to identify risks in financial transaction workflows.
- Ensure sensitive financial operations are properly authenticated, authorized, logged, and monitored.

Distributed Systems & Performance

- Build reliable microservices and distributed backend applications.
- Design asynchronous communication using Kafka, RabbitMQ, NATS, or similar messaging systems.
- Implement event-driven processing for *blockchain* and payment events.
- Improve database performance, caching strategies, and service communication.
- Design systems that handle duplicate messages, temporary failures, retries, and unexpected service interruptions.
- Implement rate limiting, timeouts, backpressure, and appropriate fault-tolerance mechanisms.
- Optimize transaction processing and API performance under high traffic.
- Work with DevOps engineers to improve service scalability and production reliability.

Cloud Infrastructure & DevOps

- Develop and maintain containerized backend services using Docker.
- Deploy applications to cloud infrastructure such as AWS, GCP, or Azure.
- Work with Kubernetes and cloud-native deployment environments.
- Contribute to CI/CD pipelines using GitHub Actions, GitLab CI, Jenkins, or similar tools.
- Implement application logging, monitoring, metrics, and distributed tracing.
- Use tools such as Prom*eth*eus, Grafana, Datadog, or OpenTelemetry to investigate performance and reliability.
- Collaborate with infrastructure engineers on production deployments, incident resolution, and operational improvements.
- Maintain clear technical documentation for services, APIs, and deployment procedures.

Testing, Security & Engineering Quality

- Write clean, maintainable, and well-tested production code.
- Develop unit, integration, and end-to-end tests for backend services.
- Create automated tests for *blockchain* integrations and transaction-processing workflows.
- Review pull requests and contribute to engineering standards.
- Identify security vulnerabilities and reliability risks in backend architecture.
- Follow secure coding practices for authentication, authorization, input validation, and API protection.
- Collaborate with security engineers on *wallet* operations, transaction authorization, and financial data protection.
- Investigate production issues and implement long-term fixes.
- Mentor less experienced engineers and share practical technical knowledge.
- Participate in technical planning, code reviews, and cross-team architecture discussions.

Required Skills & Experience

- 5+ years of professional backend or full-stack software engineering experience.
- 2+ years of hands-on experience developing *Web3* applications, *blockchain* integrations, or *cryptocurrency*-related backend services.
- S*tron*g proficiency in **Node.js and TypeScript**.
- Professional experience with **NestJS, Express.js**, or similar backend frameworks.
- S*tron*g understanding of REST APIs, asynchronous programming, and backend service architecture.
- Experience designing and maintaining production microservices or distributed backend systems.
- Practical experience integrating applications with *Ethereum* or *EVM*-compatible *blockchain* networks.
- Familiarity with *Eth*ers.js, Viem, *Web3*.js, or equivalent *blockchain* libraries.

- Understanding of *blockchain* transaction lifecycles, *smart contract* events, *token* transfers, and *wallet* interactions.
- Experience integrating backend services with *smart contract*s and *blockchain* RPC providers.
- S*trong* knowledge of PostgreSQL, Redis, and database optimization.
- Experience with asynchronous processing, messaging systems, or background job frameworks.
- Understanding of idempotency, retries, fault tolerance, and transaction consistency.
- Experience implementing secure authentication, authorization, and API communication.
- Familiarity with Docker, cloud infrastructure, and CI/CD pipelines.
- Experience writing automated tests and reviewing production code.
- Ability to independently design and deliver complex backend features.
- S*trong* problem-solving skills and experience troubleshooting production systems.
- Professional English communication skills and experience working with distributed teams.

Nice to Have (Differentiators)

- Previous experience in fintech, payment processing, digital asset platforms, *DeFi*, or *cryptocurrency* exchanges.
- Experience building *blockchain* payment systems, *wallet* infrastructure, or digital asset transaction services.
- Familiarity with *Solidity* and *smart contract* development using Hardhat or Foundry.
- Experience with Arbitrum, Optimism, Base, Polygon, zkSync, or other *EVM*-compatible networks.
- Knowledge of *blockchain* in*dex*ing, The Graph, RPC infrastructure, and event processing.
- Experience developing services supporting *token swap*s, staking, rewards, or *DeFi* protocols.
- Familiarity with cross-chain bridges, *token* transfers, and *blockchain* interoperability.
- Experience with secure transaction signing, multisignature *wallet*s, or MPC infrastructure.
- Knowledge of traditional payment gateways, KYC integrations, and financial transaction reconciliation.
- Experience with Kafka, RabbitMQ, NATS, or other messaging platforms.
- Familiarity with Go, *Rust*, Python, or C# for backend services.
- Experience designing high-availability systems and scalable financial infrastructure.
- Knowledge of Kubernetes, Terraform, and cloud-native deployment patterns.
- Experience with WebSockets, gRPC, and real-time transaction monitoring.
- Familiarity with Prom*eth*eus, Grafana, OpenTelemetry, or Datadog.
- Experience working on *blockchain* gaming, sports betting, or SocialFi applications.
- Open-source contributions to backend frameworks, *blockchain* SDKs, or *Web3* infrastructure.
- Previous experience building products in early-stage startups.

Senior-Level Expectations

As a Senior *Web3* Backend Engineer, you will be expected to:

- Independently design, develop, and maintain complex backend services.
- Take ownership of important backend components from architecture through production deployment.
- Build reliable integrations between traditional applications and *blockchain* networks.
- Understand the security and financial risks associated with transaction-processing systems.
- Make practical architecture decisions that improve scalability, performance, and maintainability.
- Establish appropriate testing, monitoring, and error-handling practices.
- Collaborate closely with *blockchain*, frontend, security, and infrastructure engineers.
- Review code and mentor less experienced engineers when appropriate.
- Identify technical problems early and propose effective solutions.
- Investigate complex production issues and implement lasting improvements.
- Maintain s*tron*g documentation and engineering quality standards.
- Remain actively involved in hands-on development and technical problem-solving.

This is a senior individual contributor position. Formal people management is not required, but technical ownership, sound engineering judgment, and collaboration are important.

Example Problems You'll Work On

1. *Blockchain* Transaction Processing

Build backend services that submit, monitor, and track *blockchain* transactions.

Handle pending confirmations, failed transactions, retries, RPC interruptions, and chain reorganizations while maintaining accurate transaction records.

2. *Cryptocurrency* Payments

Develop APIs and backend workflows supporting digital asset payments, deposits, withdrawals, transfers, and refunds.

Integrate *blockchain* operations with internal payment-processing services and financial databases.

3. *Smart Contract* Event Processing

Build services that subscribe to *smart contract* events, process *blockchain* data, and update application state.

Handle duplicate events, delayed blocks, missing data, and synchronization failures.

4. *Wallet* Infrastructure Integration

Develop backend services that integrate with *wallet* providers, custody services, or secure transaction-signing infrastructure.

Implement proper authorization, transaction validation, and monitoring.

5. Financial Reconciliation

Build reconciliation services that compare internal financial records with *blockchain* transactions and external payment providers.

Identify inconsistencies and support safe recovery from failed or interrupted operations.

6. *DeFi* Backend Services

Develop APIs and transaction-processing services supporting staking, *token swap*s, rewards, digital asset management, and other decentralized financial features.

Work with *blockchain* engineers to integrate *smart contract*s and on-chain data.

7. Gaming and Sports Betting Payments

Build backend systems that support *wallet* payments, rewards, payouts, refunds, and *blockchain*-connected financial transactions for gaming and sports betting applications.

Ensure financial operations remain reliable during high traffic and unexpected service failures.

8. Scalable Distributed Infrastructure

Improve backend services to handle increasing numbers of users, transactions, and *blockchain* events.

Optimize database queries, message-processing pipelines, caching, and service communication.

Why This Role Matters

As we establish our planned software development operations in Poland, backend engineering will be essential to connecting our financial applications with *blockchain* networks and payment infrastructure.

Our products will require secure APIs, reliable financial transaction processing, scalable distributed systems, and consistent synchronization between on-chain and off-chain services.

The Senior *Web3* Backend Engineer will help build the systems that make these applications reliable and usable in real-world environments.

This position offers meaningful ownership of backend architecture, financial transaction services, *blockchain* integrations, and production engineering.

You will work closely with experienced engineers, contribute directly to critical infrastructure, and help establish s*tron*g backend engineering practices as our team grows.

We are looking for an engineer who combines **s*tron*g Node.js and TypeScript expertise, practical *Web3* integration experience, distributed systems knowledge, and a hands-on approach to building secure financial software**.

Our goal is to build fintech and *Web3* applications where financial transactions are processed securely, accurately, and reliably.